Changes made

- Local and global costmap footprints updated to account for new robot measurements
- costmap_raw for local and global costmaps added to bag command
- BT replanning rate: 2 -> 1 Hz
- planner cache obstacle heuristic: false -> true
- Global costmap size: 30x30 -> 100x100 (origin adjusted to keep the robot at the center)
- Global costmap resolution: 2.5cm -> 4cm
- Global costmap update rate 5 Hz -> 2 Hz
- Smac lattice file regenerated for a 4cm grid resolution
- Nav2 params updated to point to the new 4cm lattice file
- IMU update rate reduced to 40 Hz, rebuilt workspace.
- BT restructured to sidestep the RoundRobinNode not resetting properly
- BT global costmap clearing removed

Other updates

Confirmed that the latency timer is 1 for ttyUSB1 with the command you gave. I'm curious what lowering the update rate to 40 will do. I made this change and rebuilt the workspace.

Bag command

ros2 bag record -o real_bag10 /diffbot_base_controller/cmd_vel /cmd_vel_raw /diffbot_base_controller/odom /odometry/filtered/local /odometry/filtered/global /global_costmap/costmap /local_costmap/costmap /velodyne_points_nav /imu/data /plan /odom_goal_pose /odom_goal_pose_rviz /tf /dynamic_joint_states /tf_static /local_costmap/published_footprint /global_costmap/published_footprint /rosout /behavior_tree_log /speed_limit /speed_governor/info /local_costmap/costmap_raw /global_costmap/costmap_raw

Course Photos

Course Description

Bit of a different layout, focusing this time on the specific zig-zag case and tight turning. The distances between the obstacle edges and the corresponding wall side were all confirmed to be under 5 feet, the IGVC minimum clearance.

The distance between bins along the length of the course was adjusted empirically. Many teleop runs were done to calibrate it to be comfortably completable by a human while remaining somewhat challenging.

After observing the first manual teleop bags, it was clear that the top of the obstacles being slightly larger in cross-sectional area was a big deal, and IGVC's obstacles would be the opposite - larger at the base while smaller at the top, slightly.

The course was adjusted after observing the first two teleop runs again to account for the obstacle's cross sectional area at the top. The bin distances to the walls were all confirmed to be between 55" and 60".

Run #1

The robot successfully completed the course. The bot was purposely positioned far away from the starting line to observe how that might affect navigation and assess speed along straights. Manual teleop mode was enabled before the bot reached the goal.

Run #2

This was a compound bag of two waypoints. The robot started very far down the hall that we've been driving in, near another lecture hall. It was given two waypoints to drive it up to the finish of the course and through it, back to the start.

I noticed that as it entered the corridor on the first waypoint, oscillations increased and the robot slowed down significantly, before entering the corridor - at which point it sped up again. The straight had a bit of oscillation, but nothing that stood out. Again, manual teleop was enabled before the goal was reached.

Once it got to the course, it completed it flawlessly. After this run, the course was adjusted to make the bins slightly closer together along the horizontal stretch, but the distance between each bin's edge and the wall was unchanged.

Run #3

Same setup as run #1 with the modified course.

Did not appear to touch any obstacles, but was going *very* slowly. Was unclear how close it got, the view was unclear.

Run #4

Driving from finish -> start. Seemed to go a bit faster this time.

Run #5

Finish -> start again, similar to run #4.

Run #6

Similar to run #2, except in the other direction. The robot was started very far away and given a single waypoint at the end of the course.

It was at this point we realized that the course may have been shifted a bit, but where exactly it got shifted was unclear. After this, the course was adjusted in a similar fashion to before: shorten the distance between bins along the length of the course while keeping the wall-to-bin clearance the same.

Run #7

Start -> finish starting at the start. The course was made harder after this again.

Run #8

Run on the much harder course. It got past the first few obstacles, but got stuck (without entering recovery) around the 3rd/4th. Once it got past those, it entered recovery, at which point… weird stuff happened. rviz showed the final obstacle on the costmap, but it kept failing to turn around it. It got into a loop, where it would approach the final obstacle, enter recovery, back up, try to approach it again, and repeat. Only once it decayed (in rviz, visually), did the robot attempt to drive to the goal, but since the costmap was gone, it hit the obstacle.

Run #9

Start -> finish, cleared the course. Amazing.

Run #10

We made the course a lot more difficult and this time the robot failed to get past the first set of bins. It went into recovery mode multiple times but was unable to maneuver around the course. It repeatedly crashed into the bin.

Report Requests

- How well did the behavior tree changes work?
- What did the loop rates look like?
- What happened in run #8?
- What happened in run #2, when the robot oscillated and slowed down as it entered the corridor?
- What specifically slows down MPPI?
- How might we make nav2 go into recovery faster, instead of sitting there for a while w/o any recovery?